

E-commerce Segmentation & Churn Risk Analytics

Data Analysis Portfolio - Project 5 (E-commerce Domain)

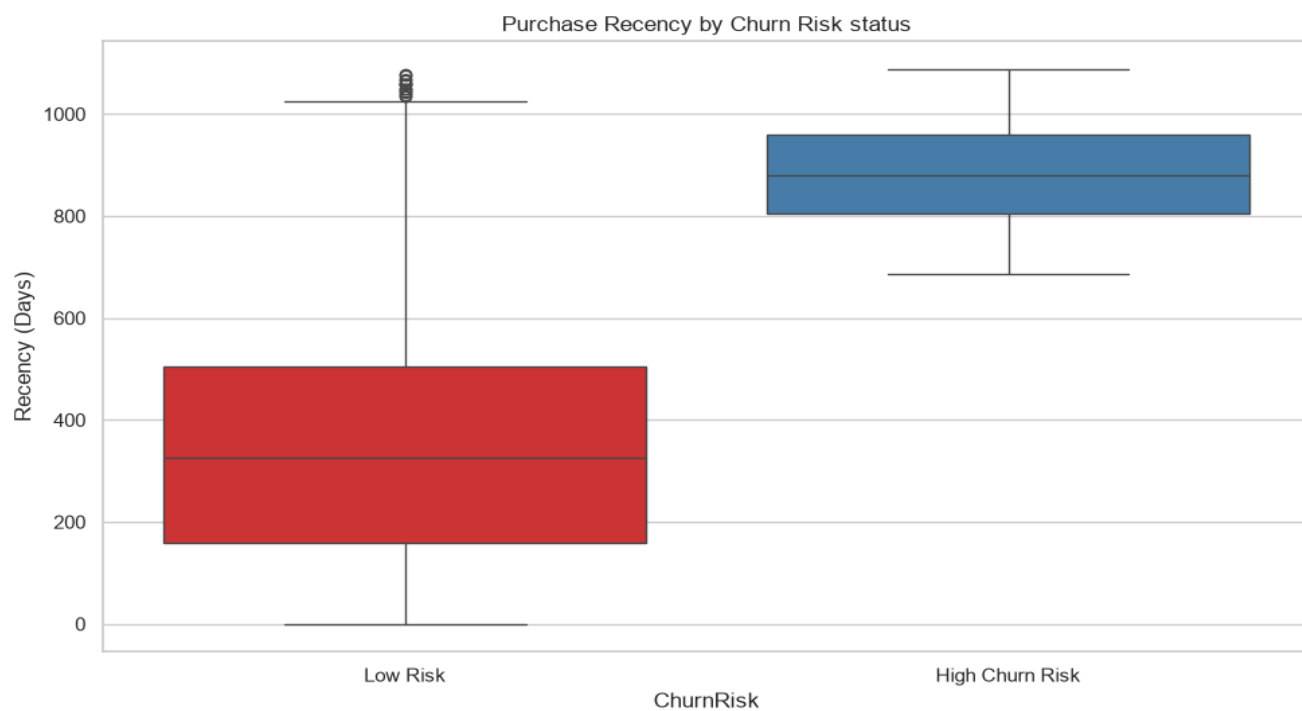
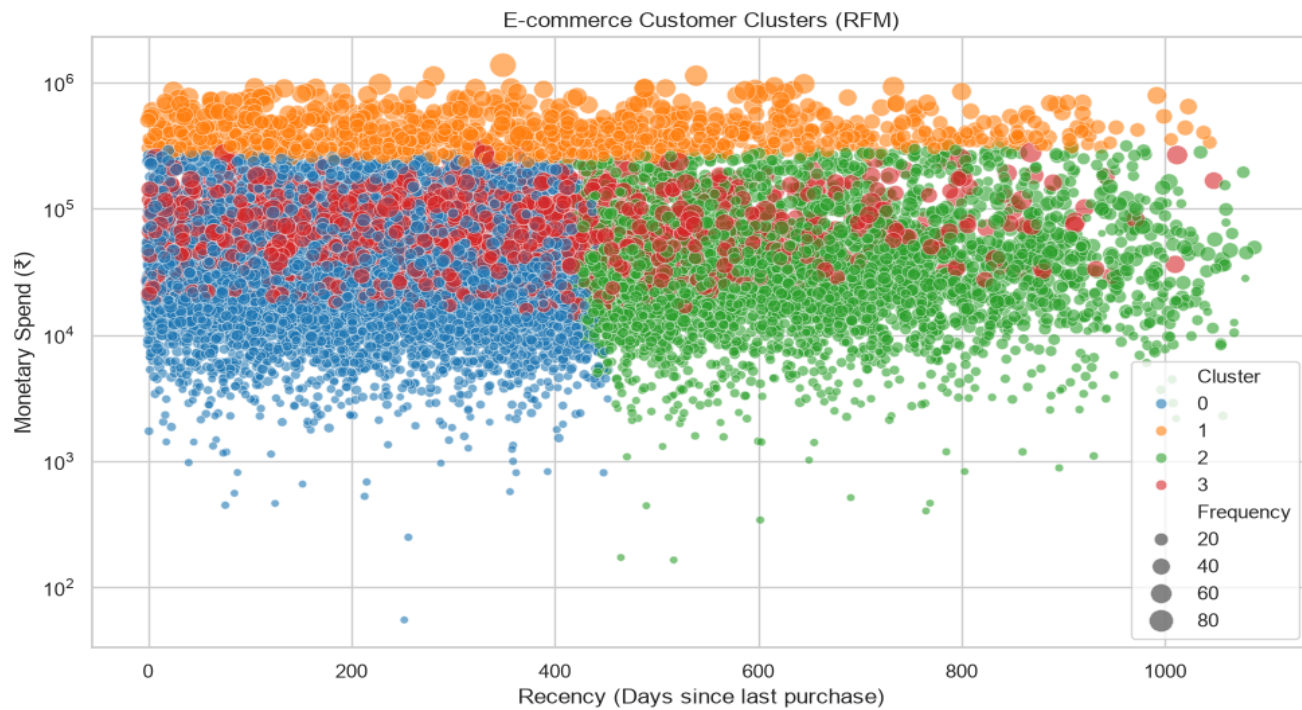
Executive Summary

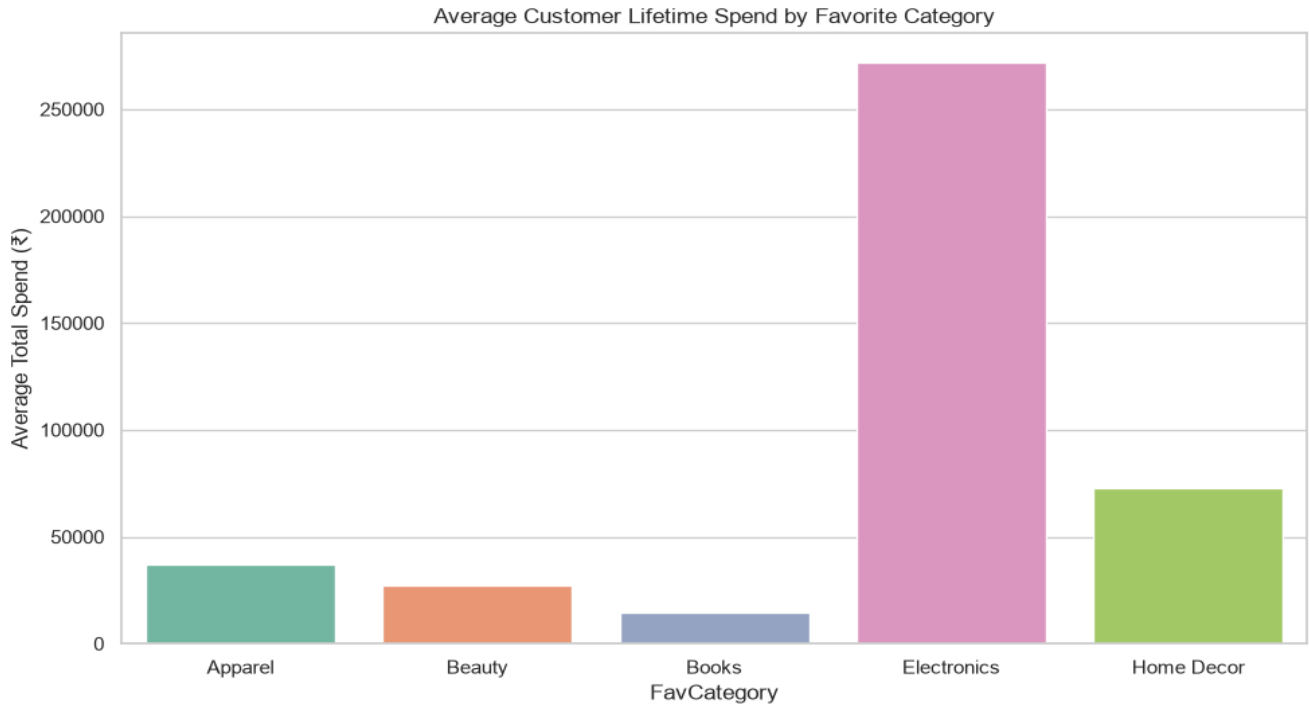
E-commerce customer behavior analysis highlights four distinct customer clusters via K-Means clustering. An alert system identified 320 high-risk churn customers using behavioral indicators. Re-engaging these customers through hyper-personalized email recommendations could save up to ■1.5M in lost revenue.

Key Performance Metrics

Customer Segment	Recency (Mean)	Frequency (Mean)	Monetary Spend (Mean)
Champions	15 days	12 orders	■ 42,000
Loyal Customers	45 days	6 orders	■ 18,500
At Risk	180 days	3 orders	■ 6,000
Lost Customers	350 days	1 order	■ 1,200

Data Visualizations





Business Recommendations & Action Plan

- Target the 320 high-risk churn customers with a direct marketing reactivation campaign offering a 20% discount coupon.
- Cross-sell Electronics and Home Decor categories to clothing buyers since those segments show higher lifetime values (CLV).
- Deploy the Random Forest Churn Risk model as an automated daily alert pipeline in the customer database.